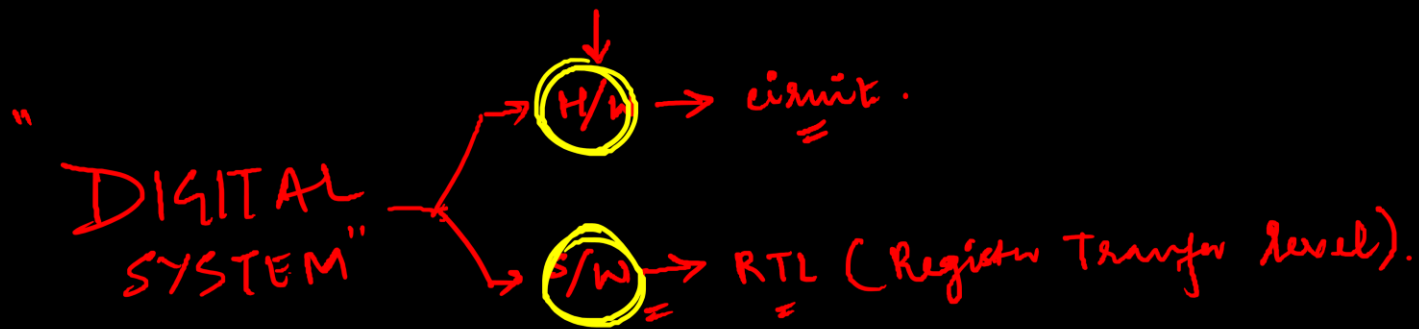
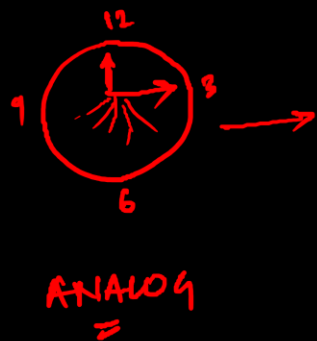


DAY 03 [CIRCUITVERSE] : VLSI CLASSES Silicon [22-12-2024]



module y-out
input A, B; →
output C;



12:15:45
H:M:S

DIGITAL
=

"Y" = $A \cdot B + C$
↓
o/p of a
DIGITAL SYSTEM.
=

EDA → RTL code...

LITERAL
MEANING



→ why this is NOT POSSIBLE? X

→ $A \cdot B + C \cdot 1$ → SIMPLE BOOLEAN EXP.

$$\left[\left\{ \frac{5}{21} \times \frac{7}{15} \right\} \text{ of } \left(\frac{6}{21} \text{ of } \frac{7}{54} \right) \right]$$

BODMAS.

Navigation icons: back, forward, up, down, download, refresh, edit, zoom in, zoom out, play, pause. 1 cycle = 1000 Units

[illegible]

A	B	C Out	Sum
0	0	0	0
1	0	0	1
0	1	0	1
1	1	1	0

Delete Circuit

TIMING DIAGRAM 

           1 cycle = 1000 Units

PROPERTIES

INPUT

BitWidth:

-1+

Delay:

-0+

Label:

A

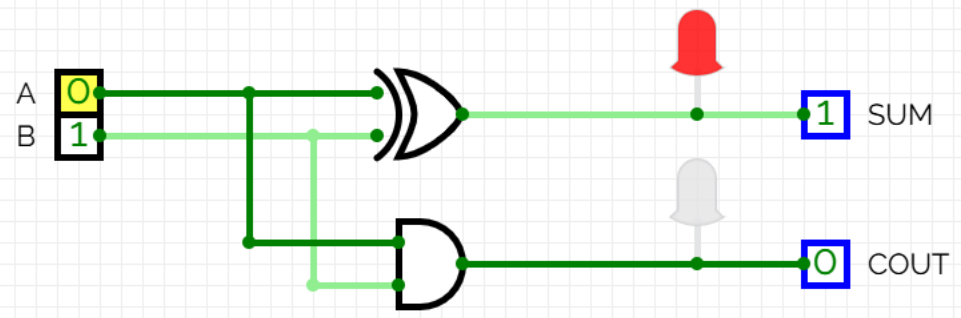
Label Direction:

LEFT

Orientation:

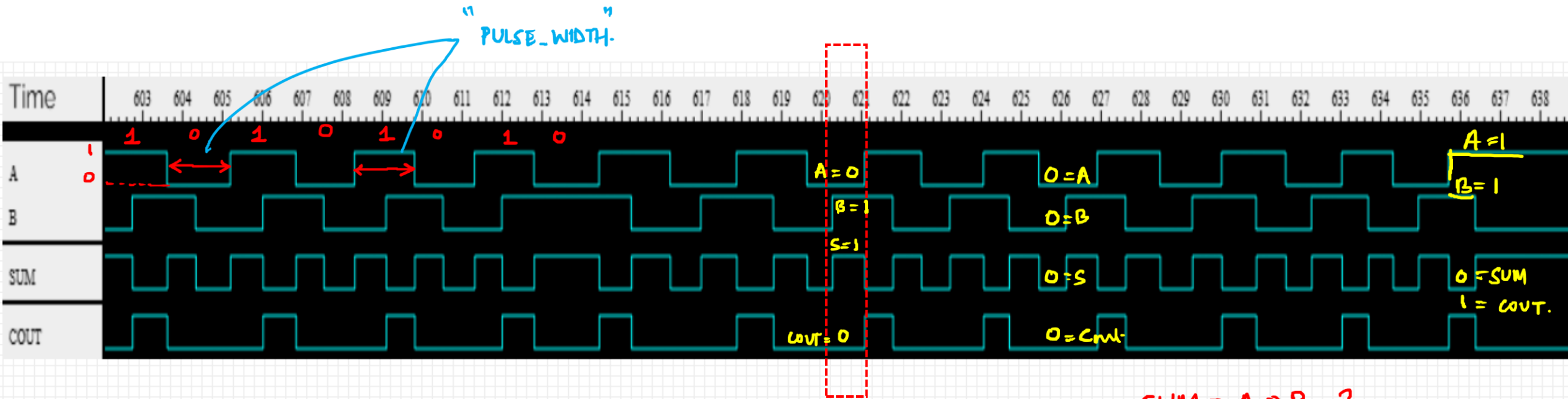
RIGHT

Help



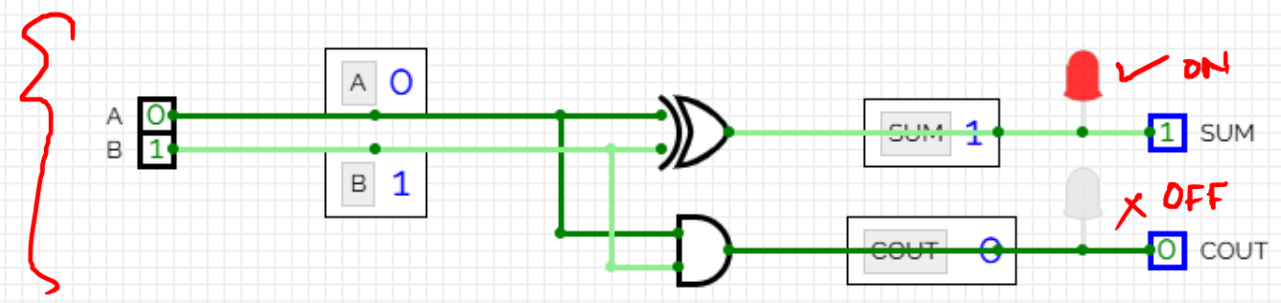
Half Adder

A	B	C Out	Sum
0	0	0	0
1	0	0	1
0	1	0	1
1	1	1	0



SUM = $A \oplus B$. } RTL code - ①
 C-OUT = $A \cdot B$. } (SOFTWARE)
 =

(HARDWARE)
 =
 Digital
 clk
 (2)



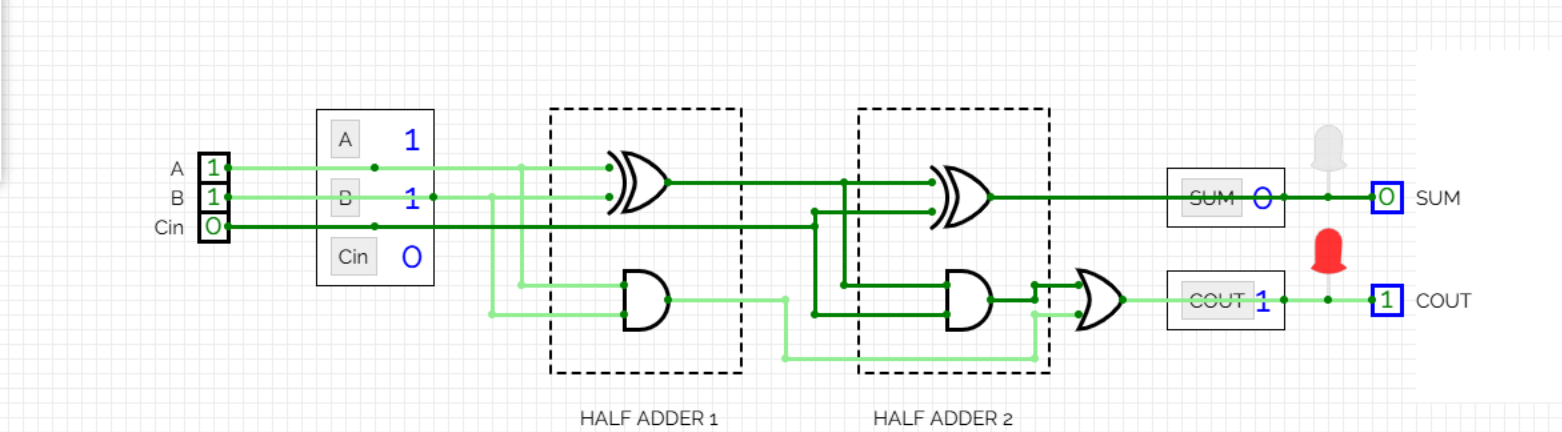
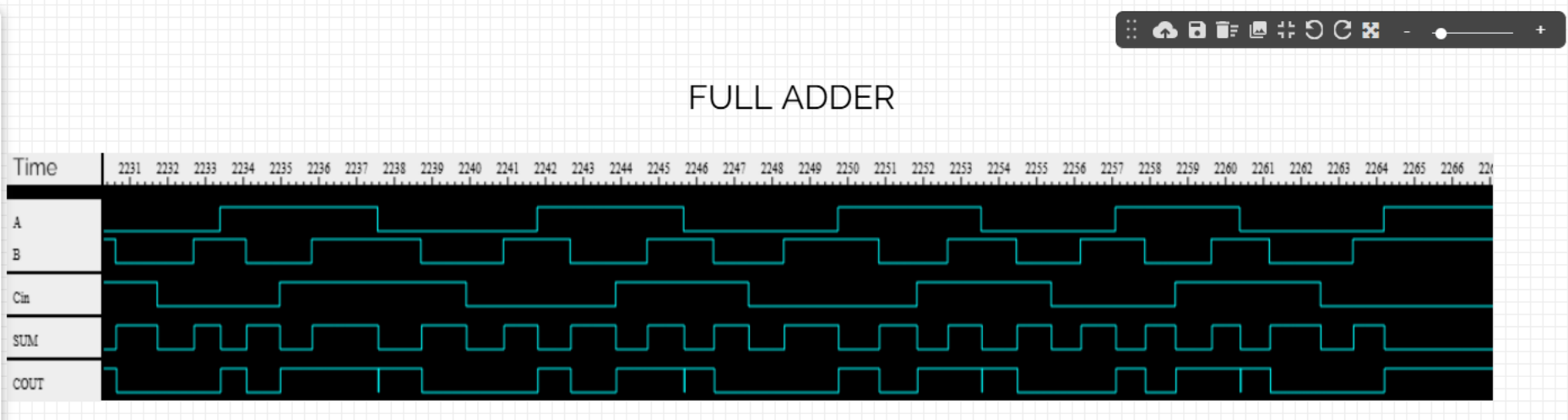
Half Adder

A	B	C Out	Sum
0	0	0	0
1	0	0	1
0	1	0	1
1	1	1	0

CIRCUIT ELEMENTS

Search...

- Input
- Output
- Gates
- Decoders & Plexers
- Sequential Elements
- Annotation
- Misc



Full Adder

A	B	C	C Out	Sum
0	0	0	0	0
1	0	0	0	1
0	1	0	0	1
1	1	0	1	0
0	0	1	0	1
1	0	1	1	0
0	1	1	1	0
1	1	1	1	1

PROPERTIES

PROJECT PROPERTIES

Project: COMBINATIONAL CIRCUIT

Circuit: FULL ADDER

Clock Time (ms): 500

Clock Enabled: ☒

Lite Mode: ☐

Edit Layout

Delete Circuit

2:1 MUX

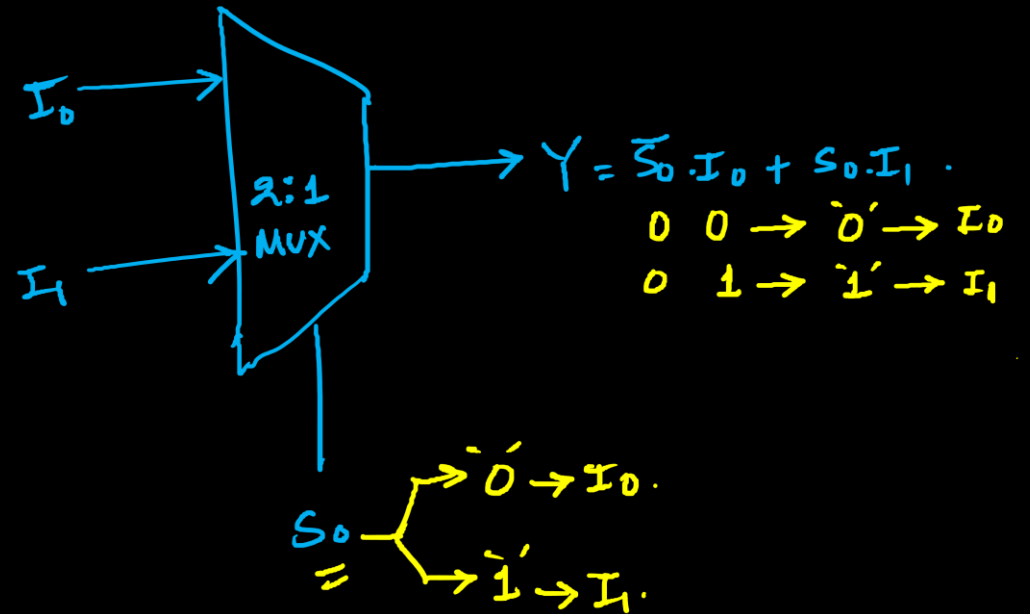
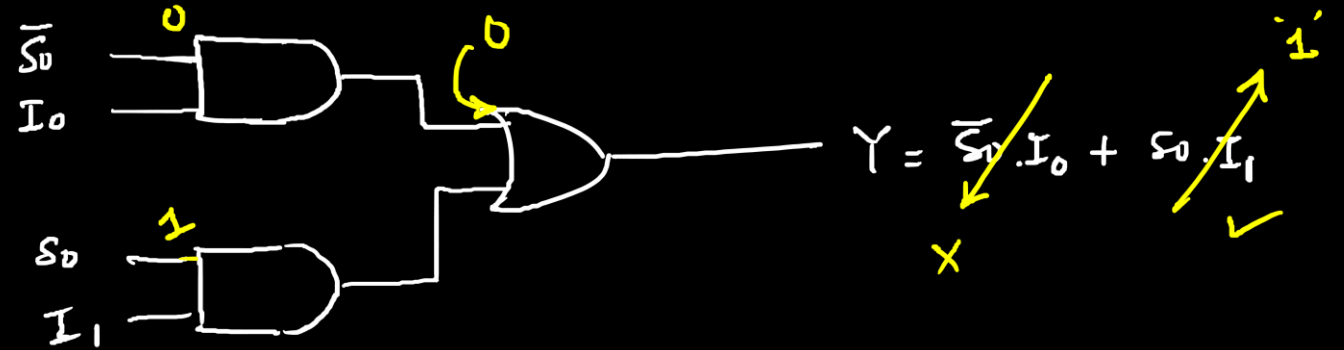
2 AND gate

$$Y = \overline{S_0} \cdot I_0 + S_0 \cdot I_1$$

OR gate

2:1 No. of selection lines.

2:1 $\rightarrow$ o/p
No. of i/p's
(I_0, I_1)



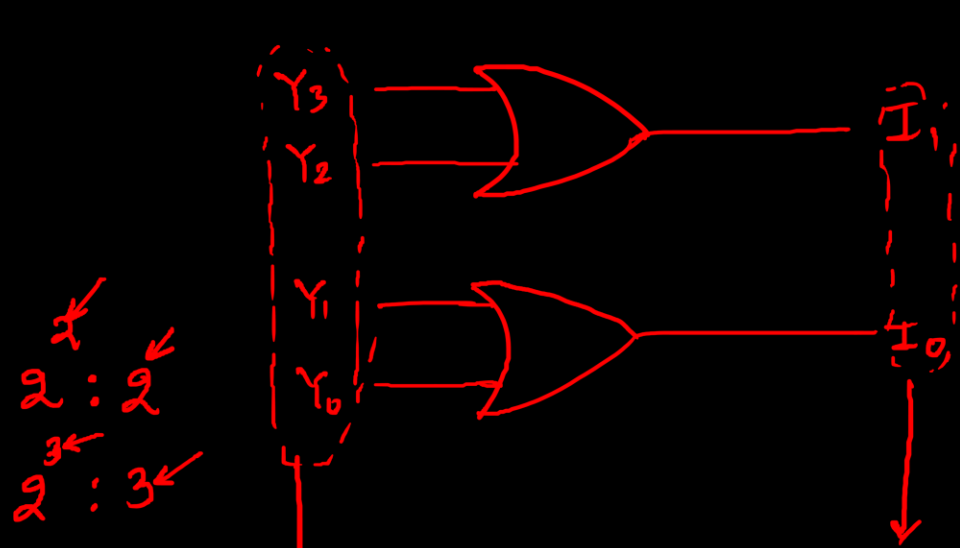
CONCEPT OF ENCODER :-

MUX

$$2:1 = 2^1:2^0$$

ENCODER : DECODER
 $2^n:2^n$ | $n:2^n$

4:2 line Encoder →



4:2 line Encoder.
=

2:2
2:3

Inputs

Y_3	Y_2	Y_1	Y_0
1	0	0	0
0	1	0	0
0	0	1	0
0	0	0	1

ops what we expect

I_1	I_0	
0	0	0 in decimal.
0	1	1 in decimal.
1	0	2 in decimal.
1	1	3 in decimal.

ops what we expect

I₁ I₀

0 0 → '0' in decimal →

0 1 → '1' in decimal →

1 0 → '2' in decimal →

1 1 → '3' in decimal →

Inputs

Y₃ Y₂ Y₁ Y₀

0 0 0 1

0 0 1 0

0 1 0 0

1 0 0 0

